SECTION 01 · FOUNDATIONS

Unsupervised Learning in the ML Ecosystem

What learning without labels means, what it can find, how to judge it, and where it powers today's systems.

Supervised vs Unsupervised | Semi-Supervised | Clustering | Anomaly Detection | Embeddings | Self-Supervised

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Laxmi Kant Tiwari · kgptalkie.com

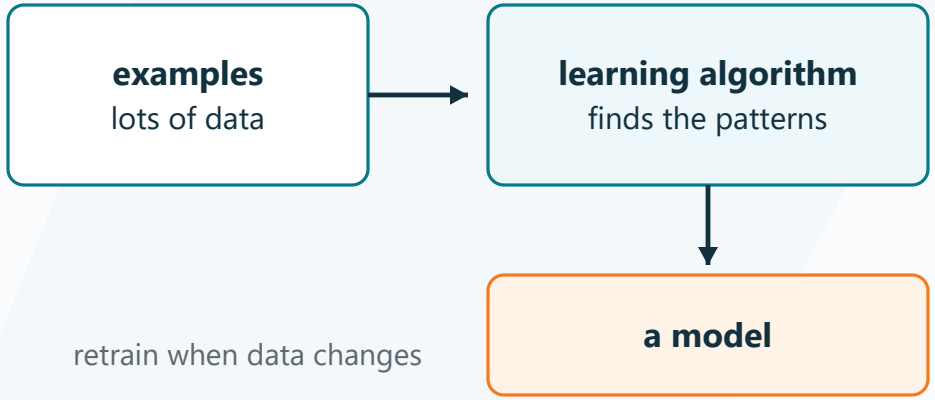
Hand-Written Rules vs Machine Learning

Rules written by hand

```
if "lottery" in text: spam
if sender not in contacts
and links > 3: spam
if "urgent" and "bank": spam
...
```

breaks when the world changes;
someone must think of every case

Patterns learned from data



retrain when data changes

Three Kinds of Machine Learning

Supervised

`x → y`

inputs with answers

learn to predict the answer

Unsupervised

`x → ?`

inputs only

learn the structure of the data

Reinforcement

`act → reward`

actions and rewards

learn what to do next

Semi-supervised learning sits between the first two: a few answers, many inputs without them.

Supervised Learning

input x	label y
email text	spam / not spam
transaction	fraud / normal
house details	price
X-ray image	disease / healthy

Classification: the label is a category

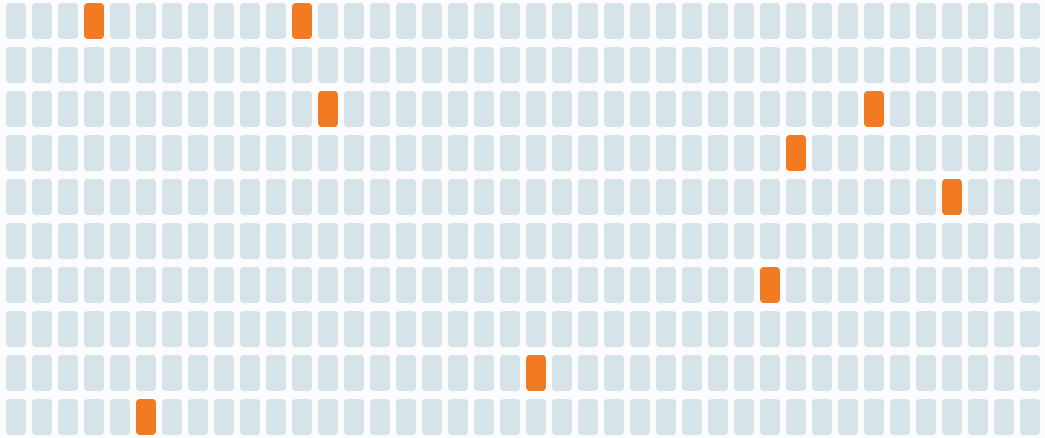
Regression: the label is a number

Strength
clear goal, accuracy you can measure

Weakness
needs many labelled examples, and only knows those

Why Labels Are Scarce

every row of data a company stores



orange: the few rows someone has labelled

Why labels are rare

- an expert must look at each example
- fraud is often confirmed weeks later
- new kinds of problems have no labels yet
- privacy rules limit who may look

Unlabelled data is cheap and huge.

Unsupervised Learning



groups	customers who buy alike
directions	a few numbers that summarise many
oddities	transactions unlike the rest
representations	vectors that capture meaning

No right answer is given, so the algorithm decides what "structure" means.

Strengths and Weaknesses

Strengths

uses data nobody has labelled
finds patterns nobody thought to ask about
adapts as new data arrives
helps supervised models as a first step

Weaknesses

no answer key: hard to say if it worked
results need a person to interpret
sensitive to scaling and settings
a group it finds may not be one you care about

Semi-Supervised Learning

six labelled points, many unlabelled



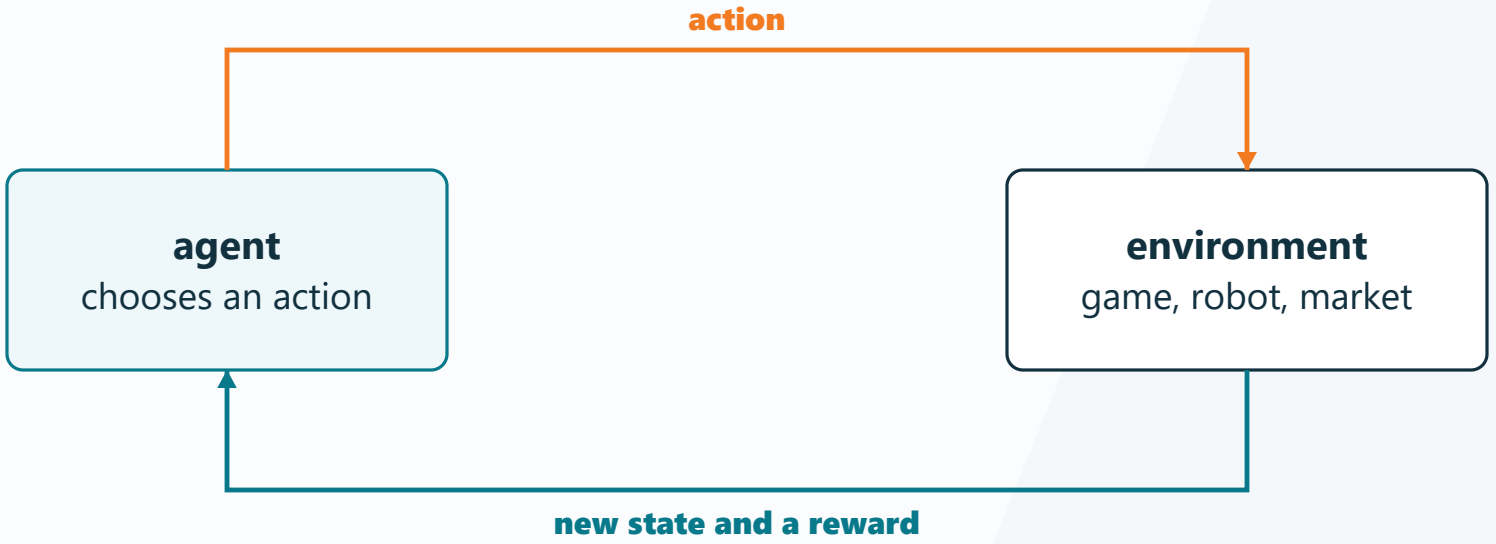
the unlabelled points still show where the groups are

A few labels plus lots of unlabelled data

- learn the structure from everything
- learn the labels from the few
- or let a model label its surest cases
- and retrain on them (self-training)

The usual situation in industry.

Reinforcement Learning



No labels, no fixed dataset:
the agent learns from
the results of its actions

Not covered in this course

Comparing the Kinds of Learning

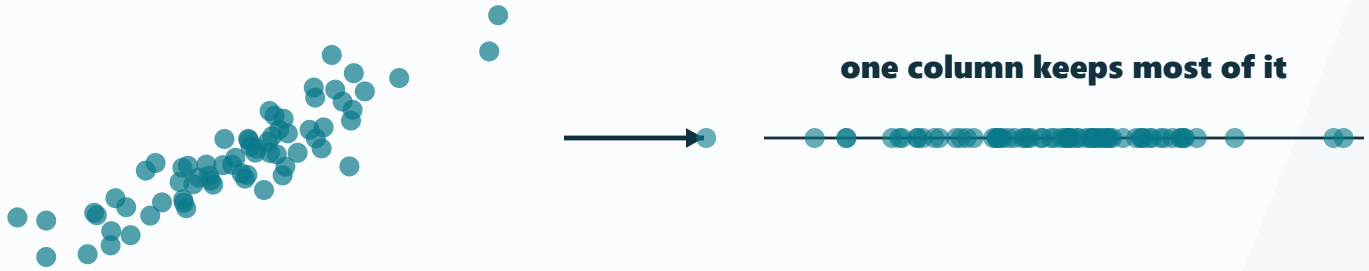
	SUPERVISED	UNSUPERVISED	SEMI-SUPERVISED	REINFORCEMENT
Data	inputs with labels	inputs only	few labels, many inputs	actions and rewards
Learns	input to label	structure in the inputs	input to label, helped by structure	which action to take
Judged by	held-out labels	usefulness, held-back labels if any	held-out labels	total reward
Examples	spam filter, price model	segments, fraud scores, embeddings	fraud with a few confirmed cases	game playing, robots

The Families of Unsupervised Learning

Dimensionality reduction	Clustering	Anomaly detection	Representation learning	Generative models
many columns to a few	points into groups	points unlike the rest	data into useful vectors	new data like the old

The families mix: reduce before clustering, cluster to find anomalies, learn vectors to do both.

Dimensionality Reduction



Why bother

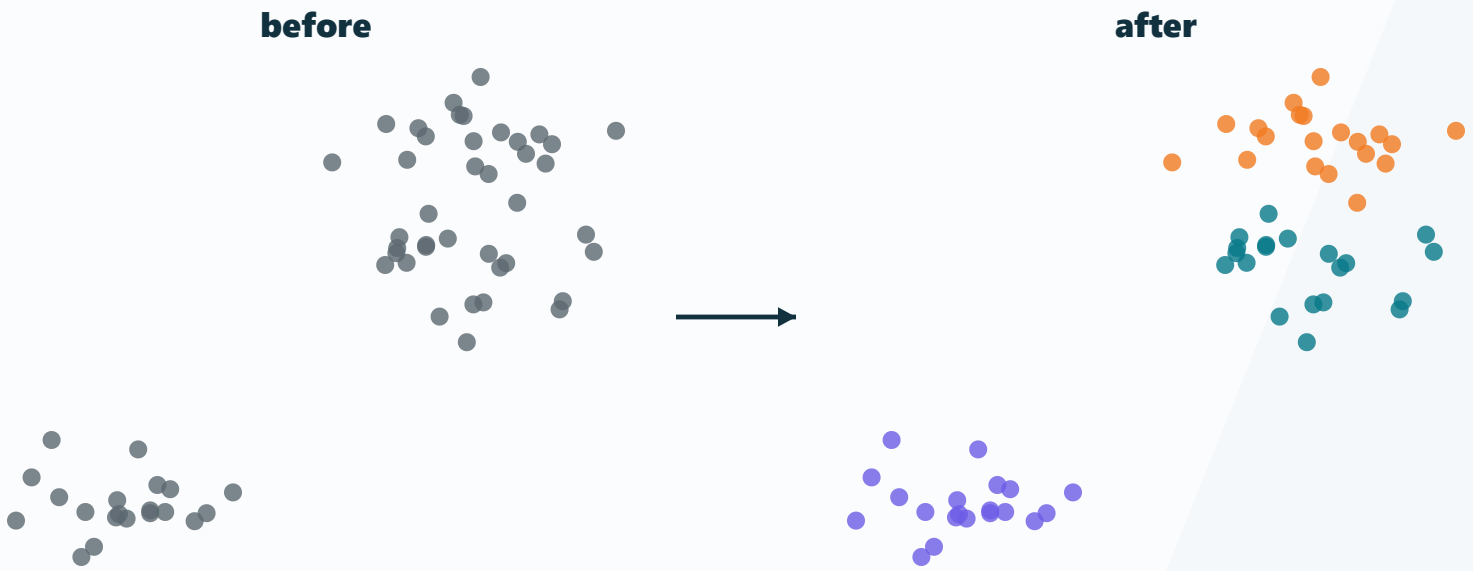
faster models

less noise

plots you can see

distances that mean something

Clustering



Uses

customer segments

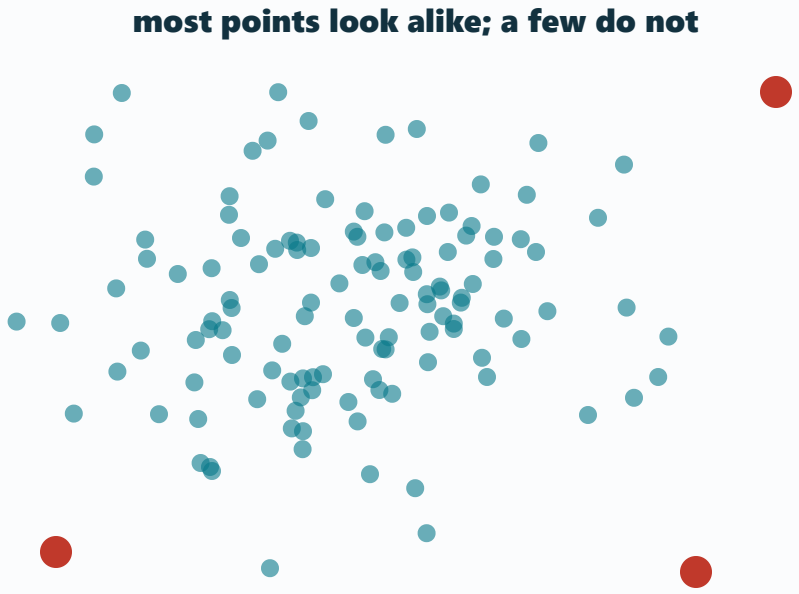
grouping documents

finding product categories

image compression

FIND THE POINTS THAT DO NOT FIT

Anomaly Detection

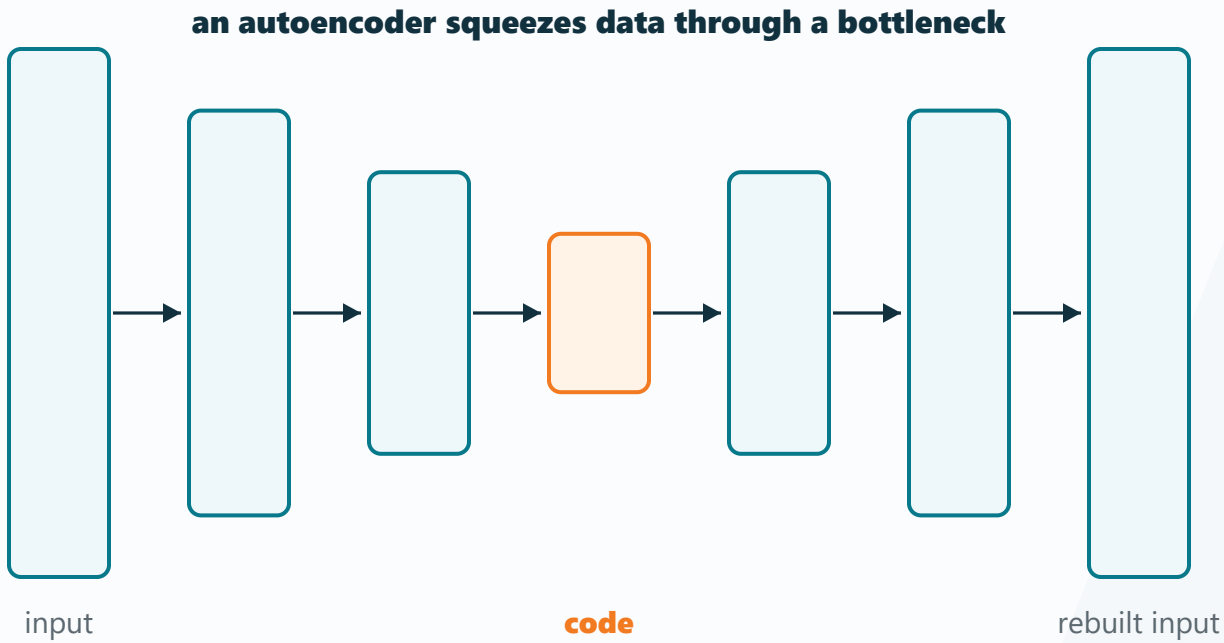


Learn what normal looks like,
then score how far each point is from it

Uses

card fraud and money laundering
machine failures from sensors
network intrusions
bad rows before training a model

Representation Learning



The network must keep what matters to rebuild the input

The code is a learned representation
use it as features, to compare items,
or its rebuild error to flag oddities

Generative Models

Learn the distribution of the data well enough to draw new samples from it

RBMs and deep belief nets

2006: pretraining layer by layer

→

GANs

2014: a generator against a judge

→

Diffusion models

2020: remove noise step by step

→

Large language models

predict the next word, at scale

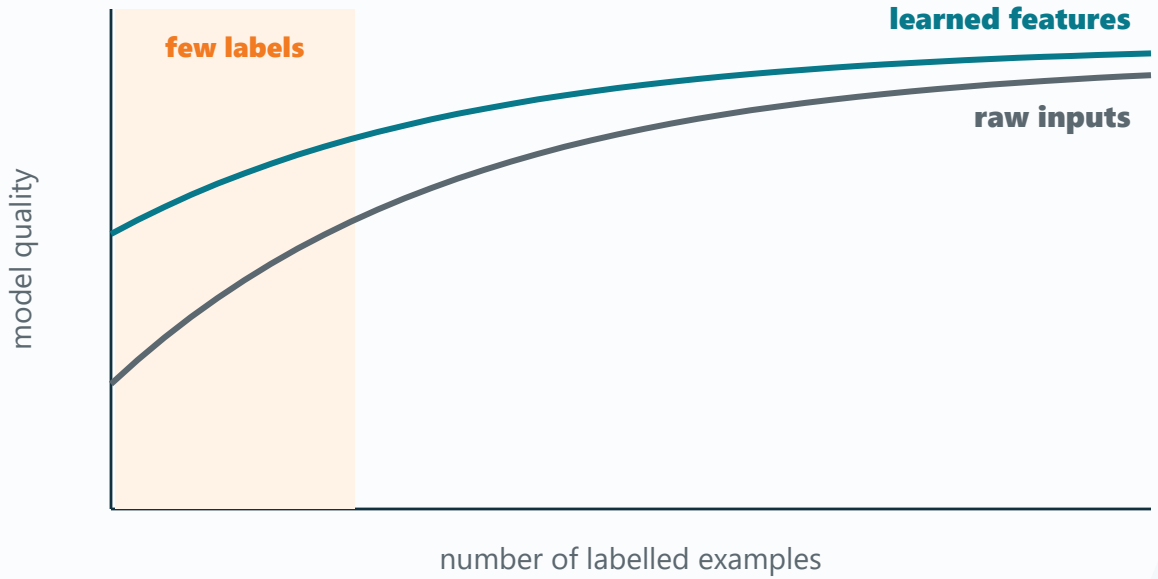
In this course: the ideas, not the models

the early ones are history; today's are large self-supervised models, used through their embeddings

Helping Supervised Projects

PROBLEM IN A SUPERVISED PROJECT	HOW UNSUPERVISED LEARNING HELPS
Too few labels	learn from all the data, then use the few labels; label the surest cases automatically
Too many columns	reduce dimensions so models train faster and distances mean something
Overfitting	fewer, smoother features leave less noise to memorise
Hand-made features	learned representations replace weeks of feature engineering
Outliers in the data	anomaly scores find bad rows before they reach training
Data drift	compare new data with old clusters or reconstruction error to spot change

Learned Features When Labels Are Few



Learn features from all the data, then train on the few labels

The gap is widest when labels are scarce and closes as labels pile up.

Shape of the idea, not a measurement
the same reason pretrained models help

How to Judge Results Without Labels

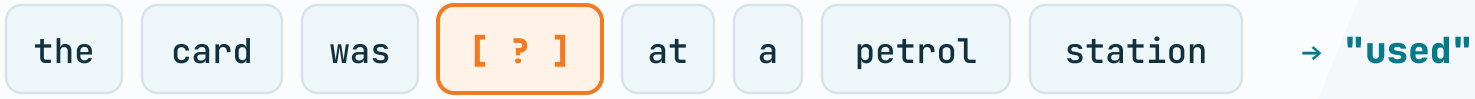
Internal scores	silhouette, inertia, reconstruction error
Stability	do the groups survive a new sample or seed?
Human review	read the top alerts, look at cluster members
Downstream task	does it make the next model or decision better?

In this course

the datasets do have labels;
we hide them while learning
and use them only at the end
to grade what was found

Self-Supervised Learning

hide part of the data and predict it from the rest



text	hide a word, predict it	language models, sentence embeddings
viewing history	hide the next item, predict it	sequence recommenders
images	hide patches, predict them	vision models

The labels come from the data itself, so no person has to write them.

Unsupervised Learning in Today's Systems

Search and RAG

embeddings find similar text

Recommenders

users and items as vectors

Fraud and monitoring

anomaly scores on live data

Segmentation

clusters for marketing

Topic discovery

clusters of messages and reviews

Foundation models

self-supervised pretraining

The Python Stack

LIBRARY	USED FOR
scikit-learn	PCA family, manifold learning, clustering, outlier detection, self-training
umap-learn	UMAP maps and features
PyOD	ECOD and other anomaly detectors
Keras on PyTorch	autoencoders in a few lines
aeon	time series datasets and k-Shape
implicit, PyTorch	recommenders from watch history
sentence-transformers, BERTopic, Ollama	text embeddings, topics, naming clusters with a local LLM

SECTION 01 · COMPLETE

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